

Course No. : CHM101
Course Title : Basic Inorganic Chemistry
Instructors : Dr. Debashis Adhikari
Co-instructor : Dr. Moitree Laskar

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Course Title : **Basic Inorganic Chemistry**
Instructors : **Dr. Debashis Adhikari and Dr. Moitree Laskar**
Further Tutoring : **by senior students**

Entry	Evaluation Component	Duration	Marks	Date Time Venue
1.	Quizzes	10 min	30-40 (20%)	In class
2.	Mid-sem II	1 hr.	30-40 (30%)	declared (9-11 th Oct)
3.	End-sem Exam.	3 hrs	60-70 (50%)	Will be declared (Nov . End)

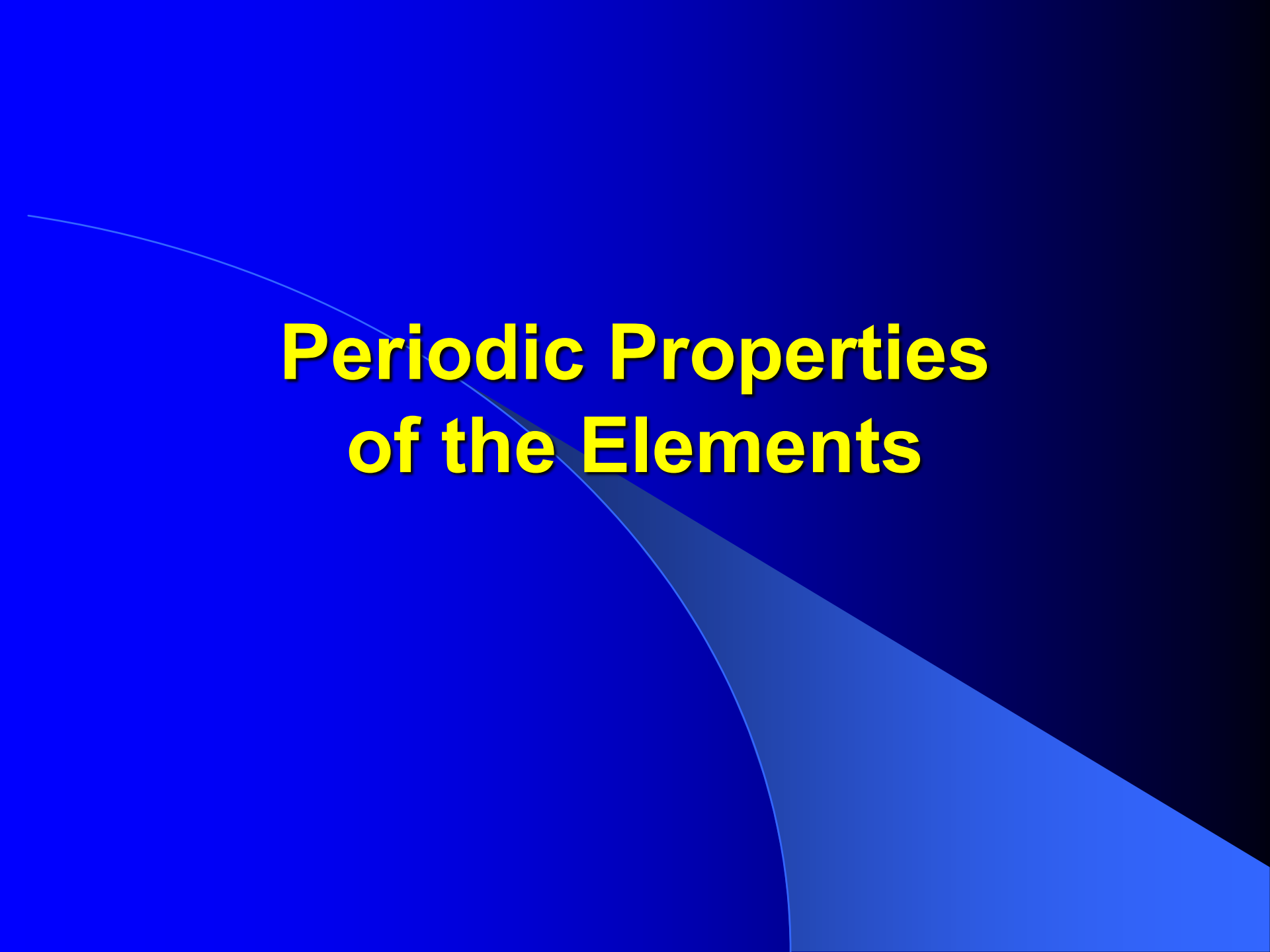
Course Outline

- Periodic table and periodicity of properties (atomic & ionic radius, electronegativity, electron affinity, ionization energy). Concept of effective nuclear charge. Electronic configuration of main group elements, transition metals, lanthanides and actinides.
- Basic concepts: HSAB principle, Polarizability, MO concept.
- Geometry of molecules: structure and bonding.
- Transition elements and their selected complexes: Structure, bonding, properties and reaction mechanism (trans effect, inner-outer sphere mechanism). Nomenclature and isomerism of coordination compounds. Theories of bonding (VBT and CFT of octahedral, tetrahedral and square planar geometries). Electronic transitions (intra-ligand, d-d and charge transfer) and magnetic behavior.
- Main group elements and their selected complexes: Group-wise discussion of s-block and p-block elements covering their oxides, oxoacids, halides and hydrides. Some unusual properties of heavy post-transition elements and their rationalization. Pseudo-halogens, interhalogens and halogen cations.

Recommended Reading

- J. D. Lee, Concise Inorganic Chemistry, Indian Edition, 5th Ed, Blackwell Science Ltd, Oxford (1996).
- F. A. Cotton, G. Wilkinson, P. L. Gaus, Basic Inorganic Chemistry, Wiley Indian Edition, 3rd Ed, John Wiley & Sons Inc, Singapore (1995).
- J. E. Huheey, E. A. Keiter, R. L. Keiter, Inorganic Chemistry, Indian Edition, 4th Ed., New Delhi.

Periodic Properties of the Elements

The background is a dark blue gradient. A light blue curved line starts from the left edge, curves downwards and to the right, and then continues as a straight line towards the bottom right corner. This line separates a lighter blue area on the left from a darker blue area on the right.

Development of Periodic Table

- 1871 – **Mendeleev** arranged the elements according to: 1. **Increasing atomic mass** 2. Elements w/ **similar properties** were placed in the **same row**
- 1913 – **Moseley** arranged the elements according to: 1. **Increasing atomic number** 2. Elements w/ **similar properties** were put in the **same column**

Development of Periodic Table

Dmitri Mendeleev and Lothar Meyer independently came to the same conclusion about how elements should be grouped.

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt									
			Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
			Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

	Ancient Times		1735–1843		1894–1918
	Middle Ages–1700		1843–1886		1923–1961
					1965–

Modern Periodic Table

The image shows a 3D periodic table of elements. The elements are arranged in rows and columns, each represented by a colored cube. The colors are: blue for noble gases, green for alkali and alkali earth metals, brown for transition metals, and grey for other elements. The table is labeled with group numbers (1A to 8A) and period numbers (1 to 7). Specific labels with arrows point to various groups: 'Alkali Metal' points to group 1A, 'Alkali Earth Metal' points to group 2A, 'Transition metals' points to groups 3B through 10B, 'Group' points to group 14A, 'Halogen' points to group 17A, and 'Noble Gas' points to group 18A. A label 'Period' with an arrow points to the first row of the table.

1 1A											13 3A	14 4A	15 5A	16 6A	17 7A	18 8A															
1 H											5 B	6 C	7 N	8 O	9 F	10 Ne															
3 Li	4 Be	5 B	6 C	7 N	8 O	9 F	10 Ne	11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar																
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr														
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe														
55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

[illegible]

TABLE 5.2Allowed Combinations of Quantum Numbers n , l , and m_l for the First Four Shells

n	l	m_l	Orbital Notation	Number of Orbitals in Subshell	Number of Orbitals in Shell
1	0	0	1s	1	1
2	0	0	2s	1	4
	1	-1, 0, +1	2p	3	
3	0	0	3s	1	9
	1	-1, 0, +1	3p	3	
	2	-2, -1, 0, +1, +2	3d	5	
	0	0	4s	1	
4	1	-1, 0, +1	4p	3	16
	2	-2, -1, 0, +1, +2	4d	5	
	3	-3, -2, -1, 0, +1, +2, +3	4f	7	

Source: web

Organization of the Modern Periodic Table

Periodic Table

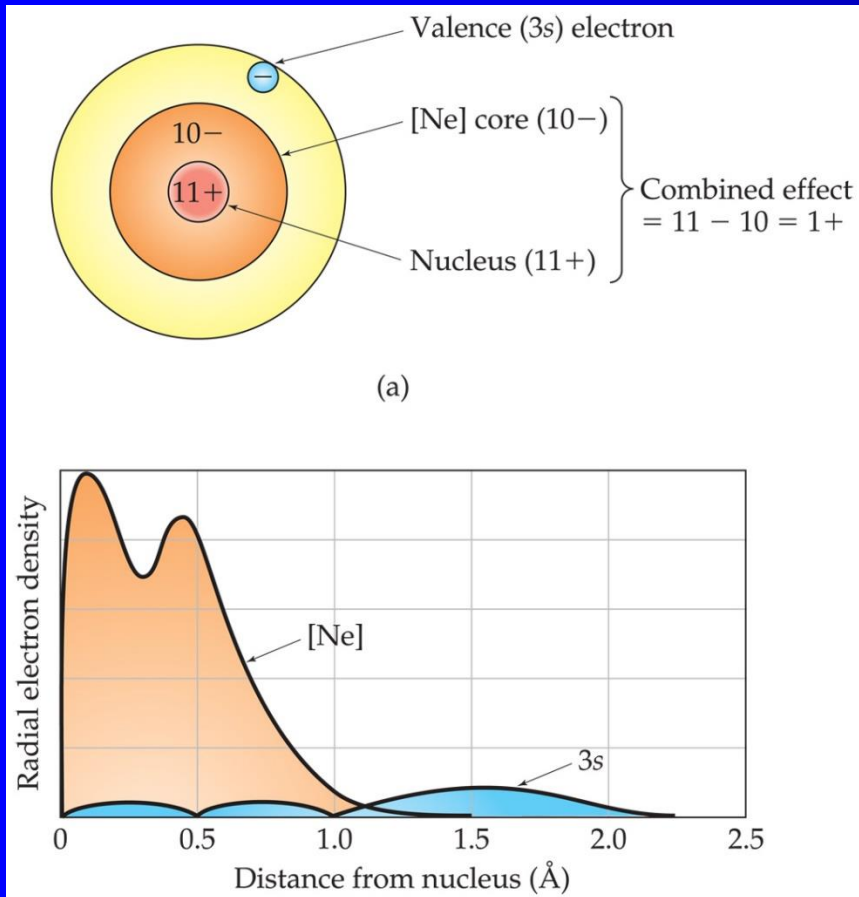
The diagram illustrates the periodic table with the following blocks and orbital filling order:

- s-block:** Groups 1A and 2A. Orbitals: $1s$, $2s$, $3s$, $4s$, $5s$, $6s$, $7s$.
- p-block:** Groups 3A to 8A. Orbitals: $2p$, $3p$, $4p$, $5p$, $6p$.
- d-block:** Groups 3B to 10. Orbitals: $3d$, $4d$, $5d$, $6d$.
- f-block:** Groups 14 and 15. Orbitals: $4f$, $5f$.

Periodic Trends

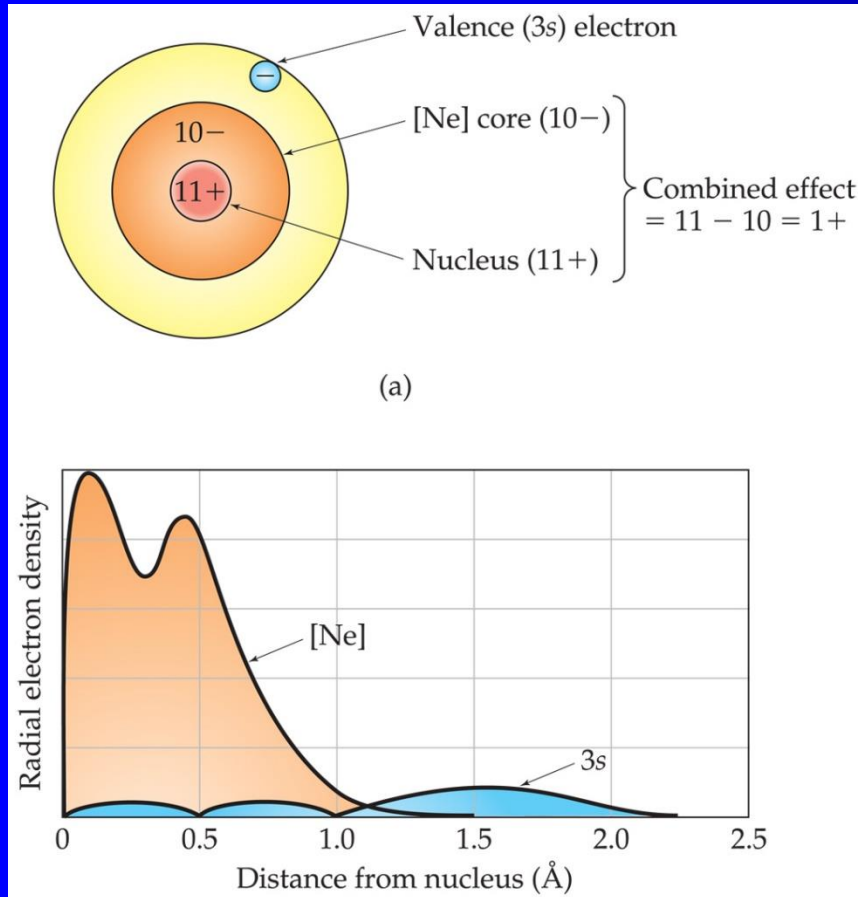
- We will rationalize observed trends in
 - **Sizes of atoms and ions.**
 - **Ionization energy.**
 - **Electron affinity.**
 - **Electronegativity.**

Effective Nuclear Charge



- In a many-electron atom, electrons are both attracted to the nucleus and repelled by other electrons.
- The nuclear charge that an electron experiences depends on both factors.

Effective Nuclear Charge

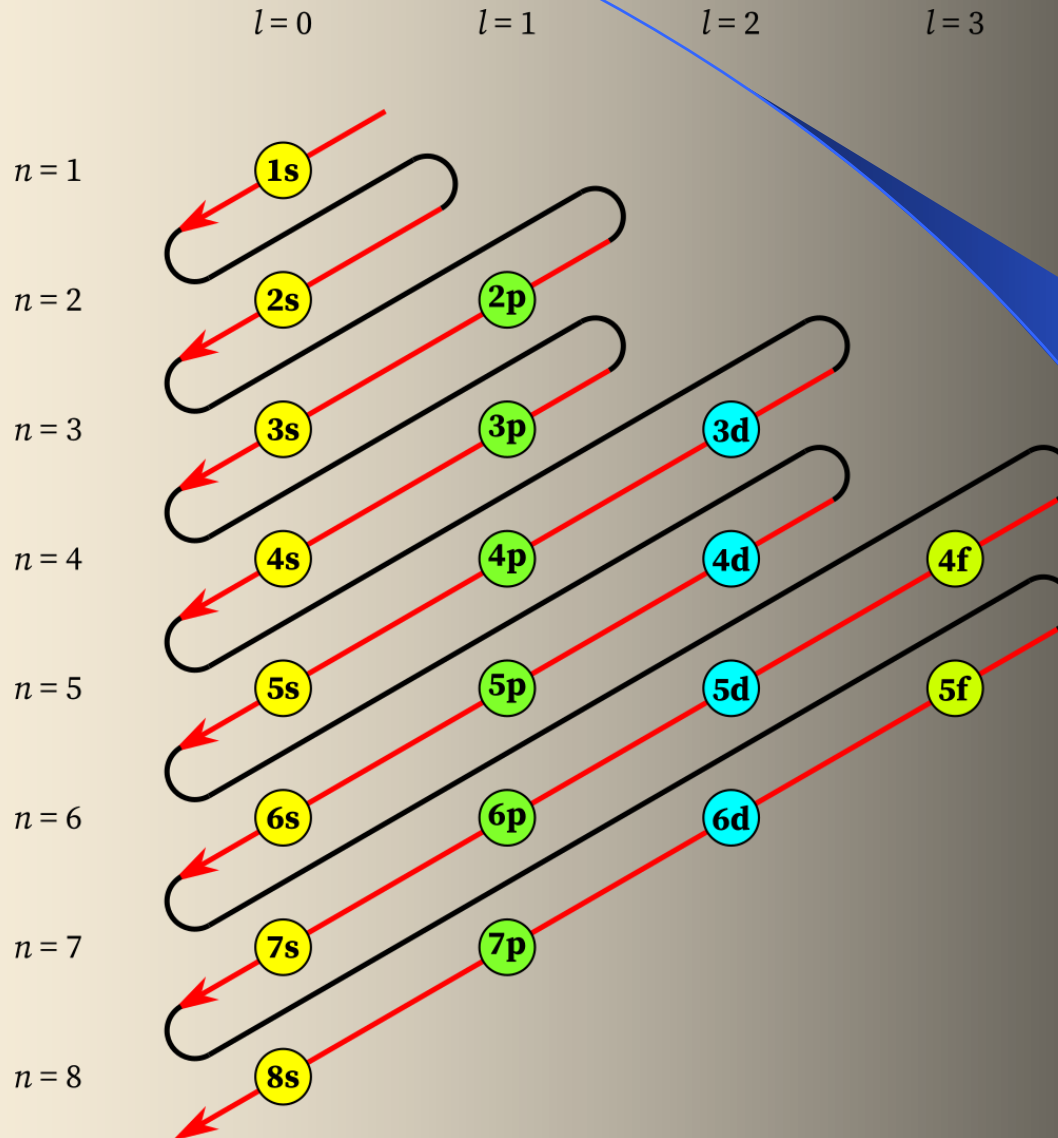


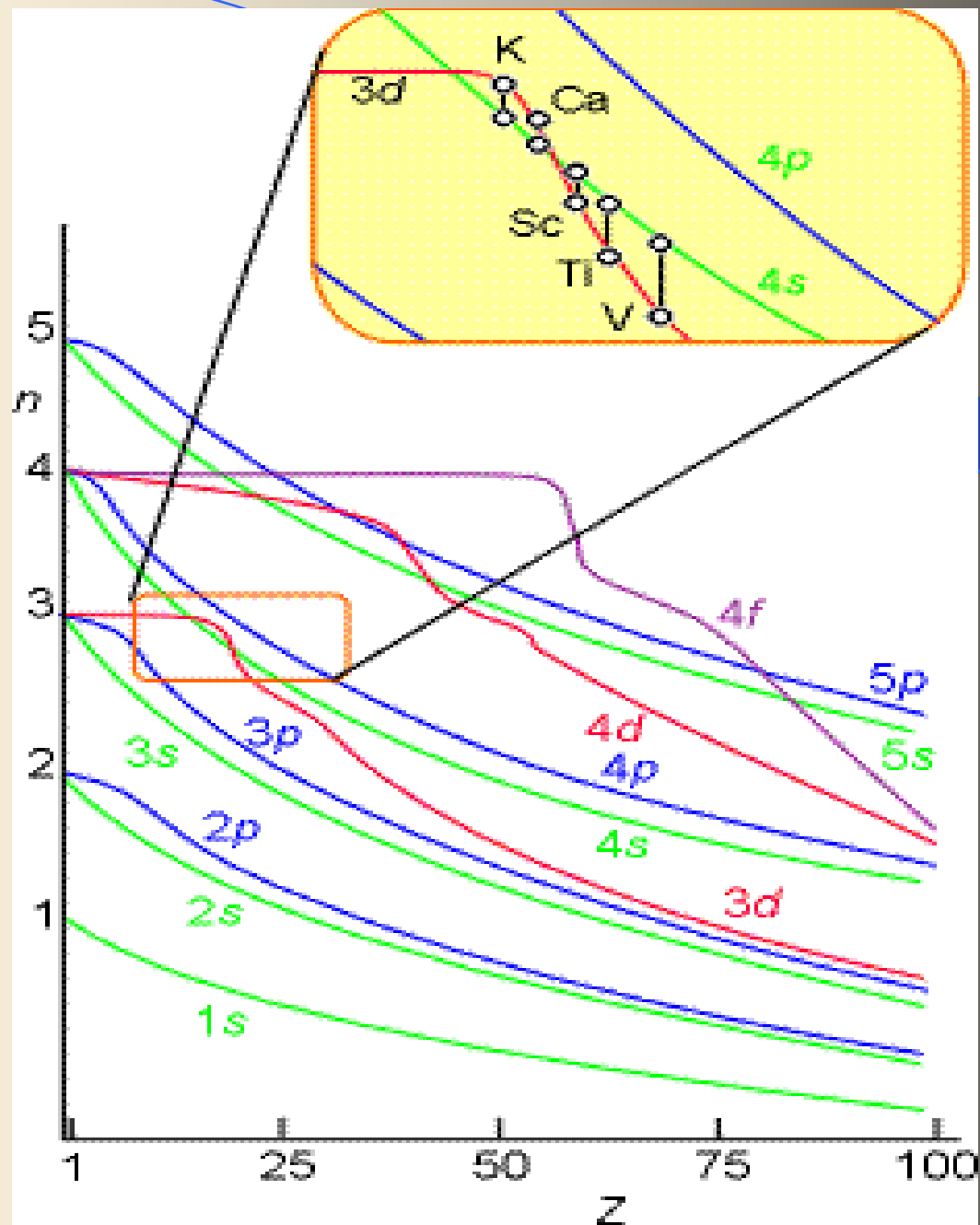
The effective nuclear charge, Z_{eff} , is found this way:

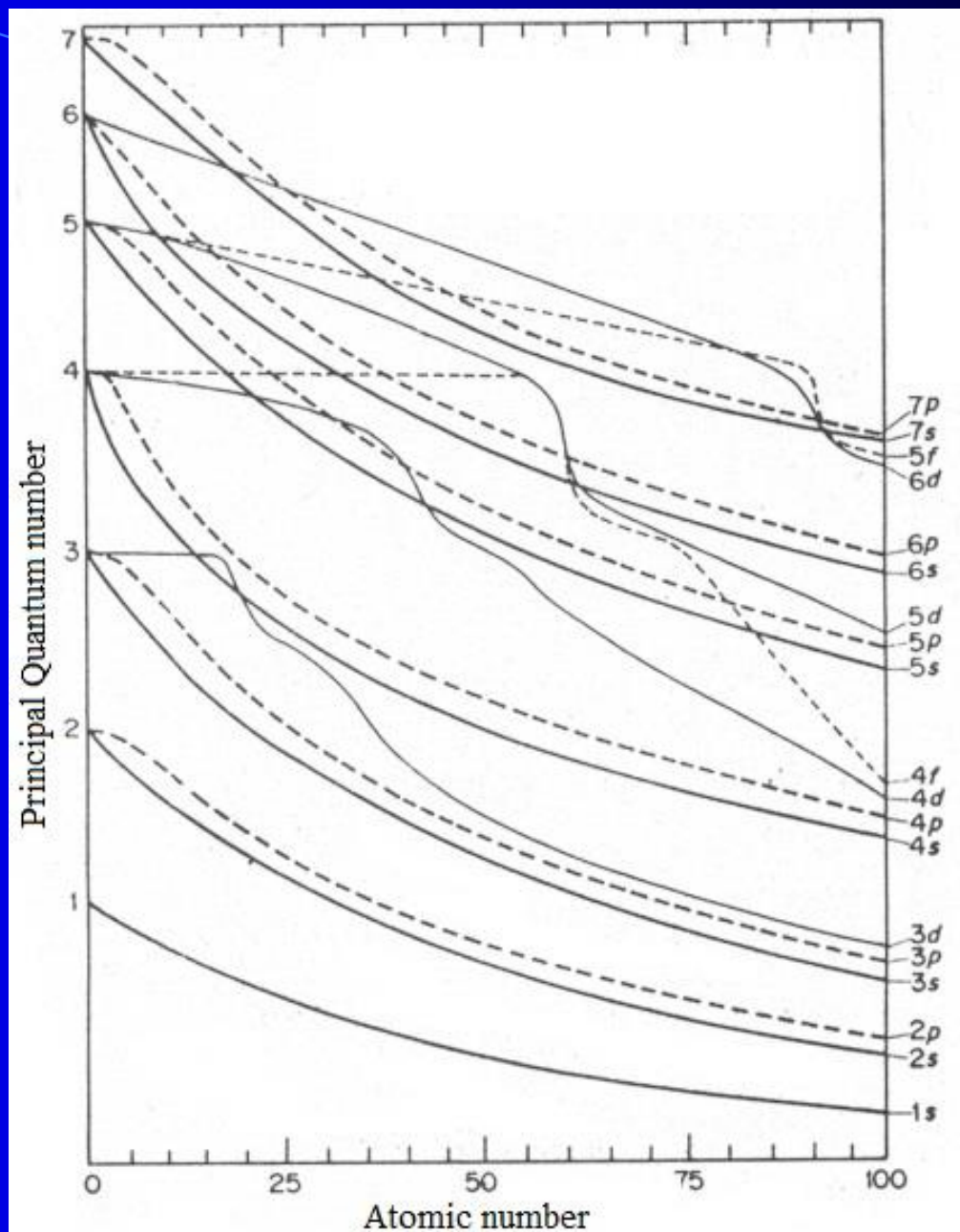
$$Z_{\text{eff}} = Z - \sigma$$

where Z is the atomic number and σ is a screening constant, usually close to the number of inner electrons.

Aufbau Principle







How to Calculate the Effective Nuclear Charge (Z_{eff} or Z^*)

If the electron resides in *s* or *p* orbital

1. Electrons in principal shell higher than the e- in question contribute 0 to σ
2. Each electron in the same principal shell contribute 0.35 to σ
3. Electrons in (n-1) shell each contribute 0.85 to σ
4. Electrons in deeper shell each contribute 1.00 to σ

Example: Calculate the Z^* for the 2p electron

Fluorine ($Z = 9$) $1s^2 2s^2 2p^5$

Screening constant for one of the outer electron (2p):

6 (six) (two 2s e- and four 2p e-) = $6 \times 0.35 = 2.10$

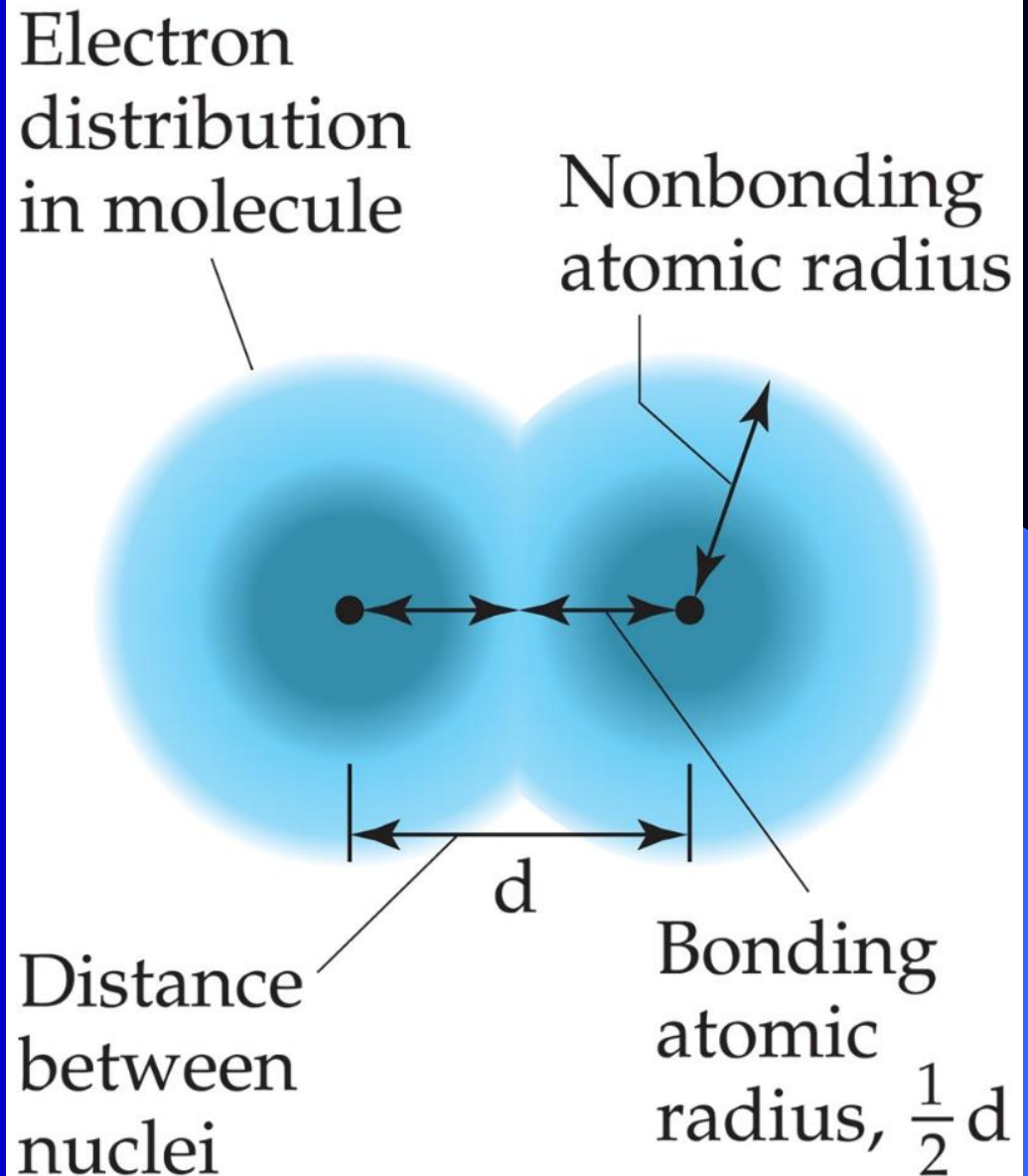
2 (two) 1s e- = $2 \times 0.85 = 1.70$

$\sigma = 1.70 + 2.10 = 3.80$

$Z^* = 9 - 3.80 = 5.20$

Sizes of Atoms

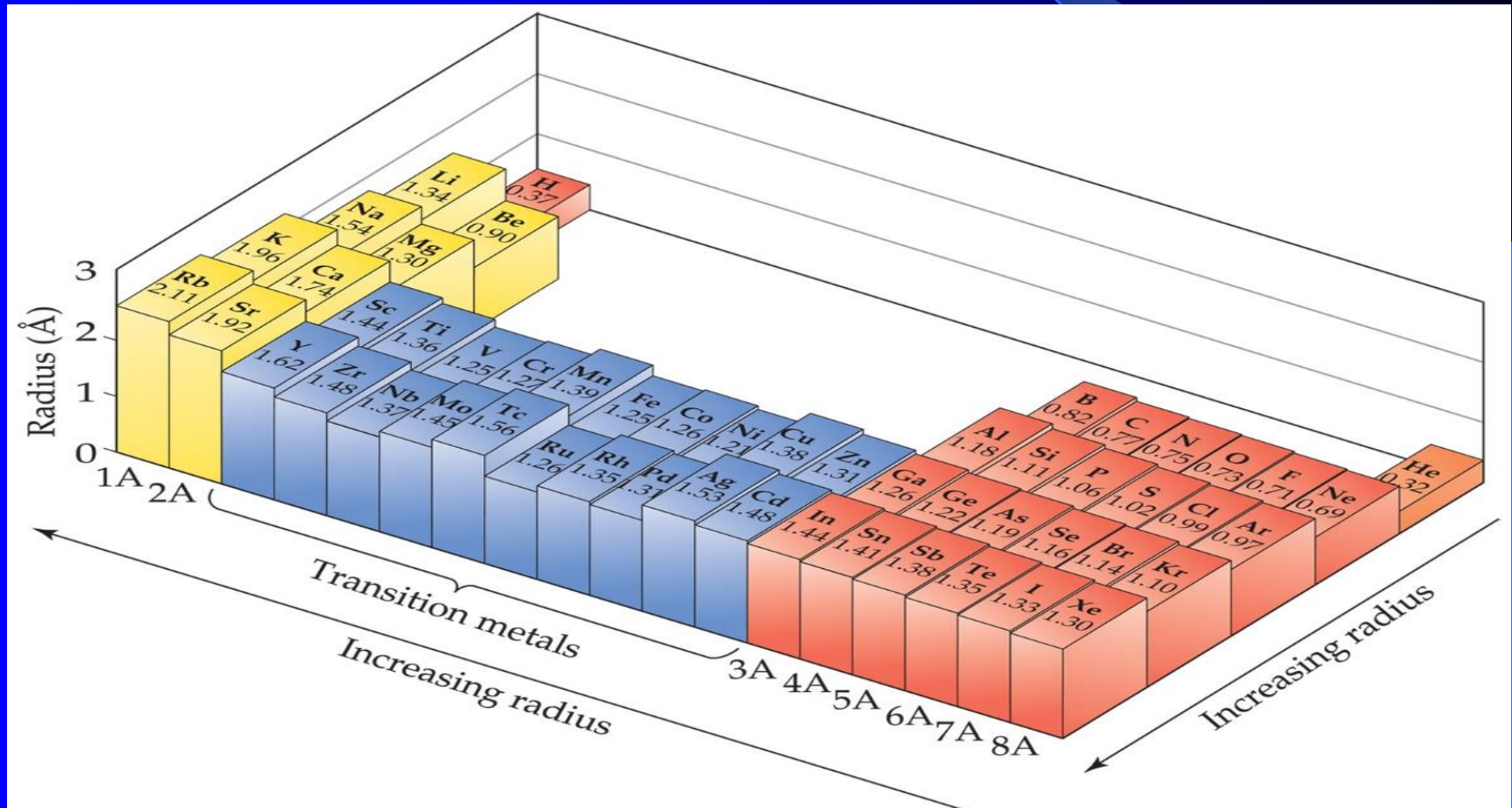
The bonding atomic radius is defined as one-half of the distance between covalently bonded nuclei.



Sizes of Atoms

Bonding atomic radius tends to

- decrease from left to right across a row due to increasing Z_{eff}
- increase from top to bottom of a column due to increasing value of n



Sizes of Ions

Group 1A		Group 2A		Group 3A		Group 6A		Group 7A	
Li ⁺	Li	Be ²⁺	Be	B ³⁺	B	O	O ²⁻	F	F ⁻
0.68	1.34	0.31	0.90	0.23	0.82	0.73	1.40	0.71	1.33
Na ⁺	Na	Mg ²⁺	Mg	Al ³⁺	Al	S	S ²⁻	Cl	Cl ⁻
0.97	1.54	0.66	1.30	0.51	1.18	1.02	1.84	0.99	1.81
K ⁺	K	Ca ²⁺	Ca	Ga ³⁺	Ga	Se	Se ²⁻	Br	Br ⁻
1.33	1.96	0.99	1.74	0.62	1.26	1.16	1.98	1.14	1.96
Rb ⁺	Rb	Sr ²⁺	Sr	In ³⁺	In	Te	Te ²⁻	I	I ⁻
1.47	2.11	1.13	1.92	0.81	1.44	1.35	2.21	1.33	2.20

• Ionic size depends upon:

- Nuclear charge.
- Number of electrons.
- Orbitals in which electrons reside.

Sizes of Ions

Group 1A		Group 2A		Group 3A		Group 6A		Group 7A	
Li^+	Li	Be^{2+}	Be	B^{3+}	B	O	O^{2-}	F	F^-
0.68	1.34	0.31	0.90	0.23	0.82	0.73	1.40	0.71	1.33
Na^+	Na	Mg^{2+}	Mg	Al^{3+}	Al	S	S^{2-}	Cl	Cl^-
0.97	1.54	0.66	1.30	0.51	1.18	1.02	1.84	0.99	1.81
K^+	K	Ca^{2+}	Ca	Ga^{3+}	Ga	Se	Se^{2-}	Br	Br^-
1.33	1.96	0.99	1.74	0.62	1.26	1.16	1.98	1.14	1.96
Rb^+	Rb	Sr^{2+}	Sr	In^{3+}	In	Te	Te^{2-}	I	I^-
1.47	2.11	1.13	1.92	0.81	1.44	1.35	2.21	1.33	2.20

- **Cations are smaller than their parent atoms.**

- The outermost electron is removed and repulsions are reduced.

Sizes of Ions

Group 1A		Group 2A		Group 3A		Group 6A		Group 7A	
Li ⁺	Li	Be ²⁺	Be	B ³⁺	B	O	O ²⁻	F	F ⁻
0.68	1.34	0.31	0.90	0.23	0.82	0.73	1.40	0.71	1.33
Na ⁺	Na	Mg ²⁺	Mg	Al ³⁺	Al	S	S ²⁻	Cl	Cl ⁻
0.97	1.54	0.66	1.30	0.51	1.18	1.02	1.84	0.99	1.81
K ⁺	K	Ca ²⁺	Ca	Ga ³⁺	Ga	Se	Se ²⁻	Br	Br ⁻
1.33	1.96	0.99	1.74	0.62	1.26	1.16	1.98	1.14	1.96
Rb ⁺	Rb	Sr ²⁺	Sr	In ³⁺	In	Te	Te ²⁻	I	I ⁻
1.47	2.11	1.13	1.92	0.81	1.44	1.35	2.21	1.33	2.20

- **Anions are larger than their parent atoms.**

- **Electrons are added and repulsions are increased.**

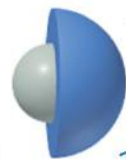
Sizes of Ions

- Ions increase in size as you go down a column.
 - Due to increasing value of n .

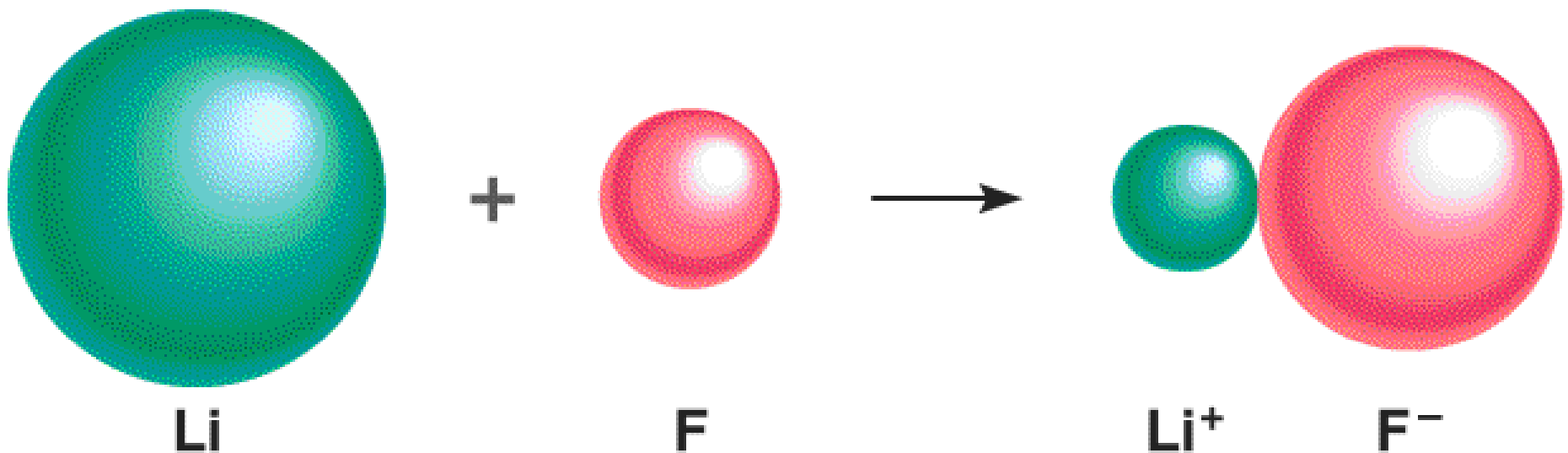
Group 3A		Group 6A	
B^{3+}	B	O	O^{2-}
0.23	0.82	0.73	1.40
Al^{3+}	Al	S	S^{2-}
0.51	1.18	1.02	1.84
Ga^{3+}	Ga	Se	Se^{2-}
0.62	1.26	1.16	1.98
In^{3+}	In	Te	Te^{2-}
0.81	1.44	1.35	2.21

Sizes of Ions

- In an **isoelectronic series**, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.

Group 1A		Group 2A		Group 3A		Group 6A		Group 7A	
Li^+	Li	Be^{2+}	Be	B^{3+}	B	O	O^{2-}	F	F^-
									
0.68	1.34	0.31	0.90	0.23	0.82	0.73	1.40	0.71	1.33

Sizes of Ions



Cation is always **smaller** than atom from which it is formed.

Anion is always **larger** than atom from which it is formed.

Ionization Energy

- Amount of energy required to remove an electron from the ground state of a gaseous atom or ion.
 - First ionization energy is that energy required to remove first electron.
 - Second ionization energy is that energy required to remove second electron, etc.

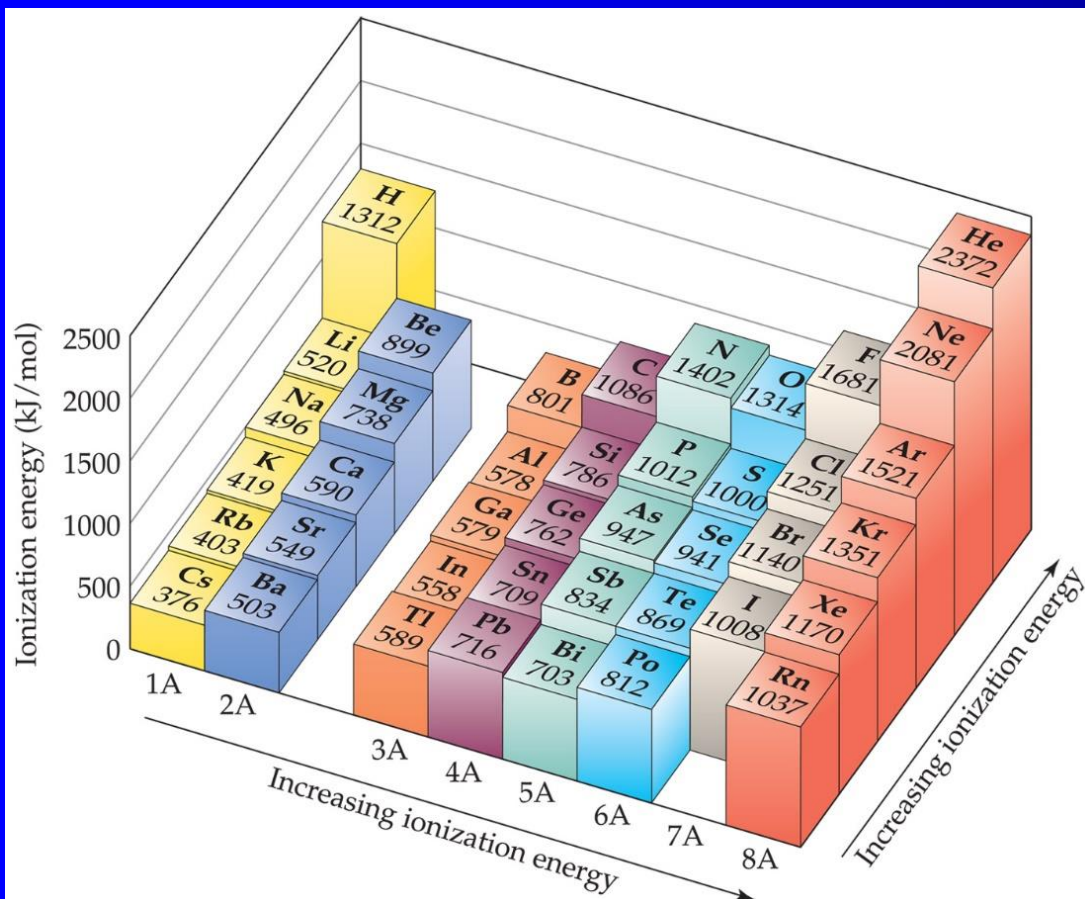
Ionization Energy

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, the ionization energy takes a quantum jump.

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

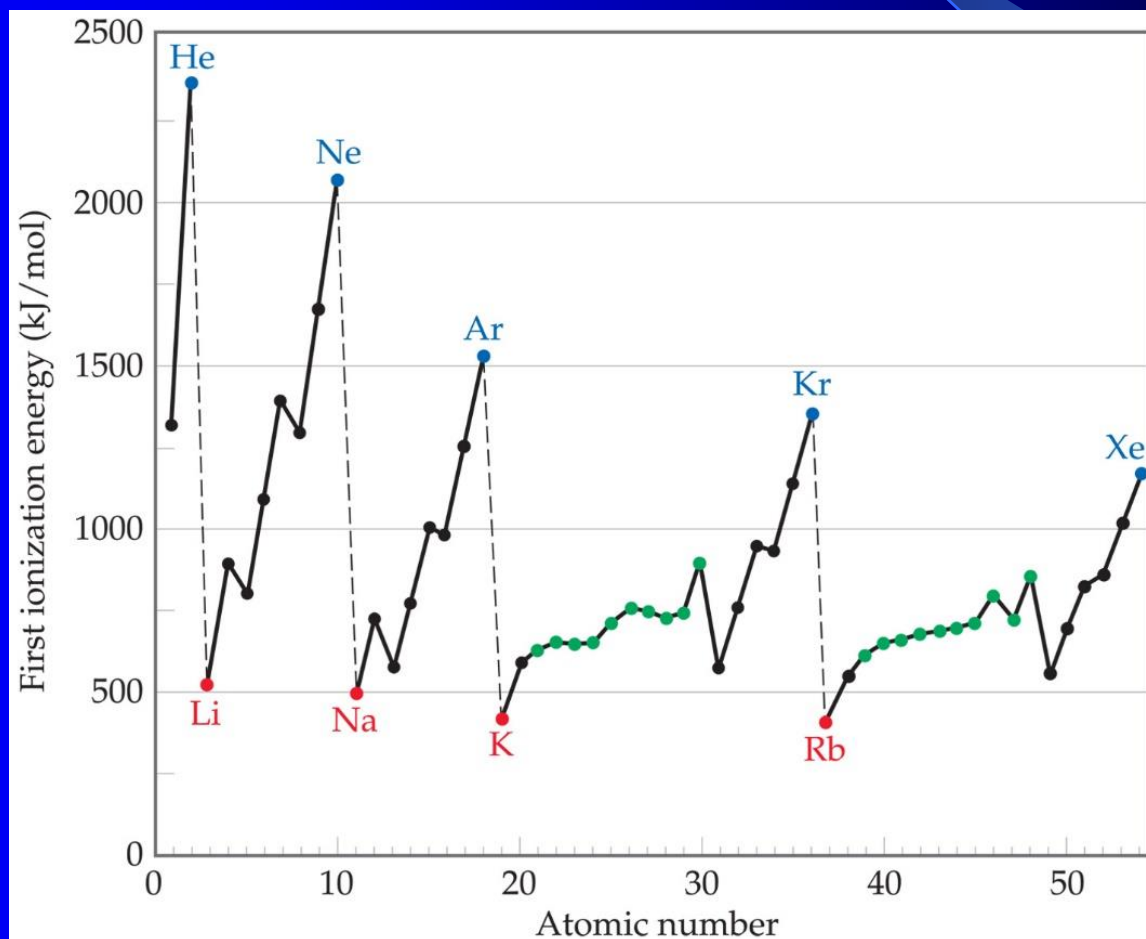
Trends in First Ionization Energies

- As one goes down a column, less energy is required to remove the first electron.
 - For atoms in the same group, Z_{eff} is essentially the same, but the valence electrons are farther from the nucleus.



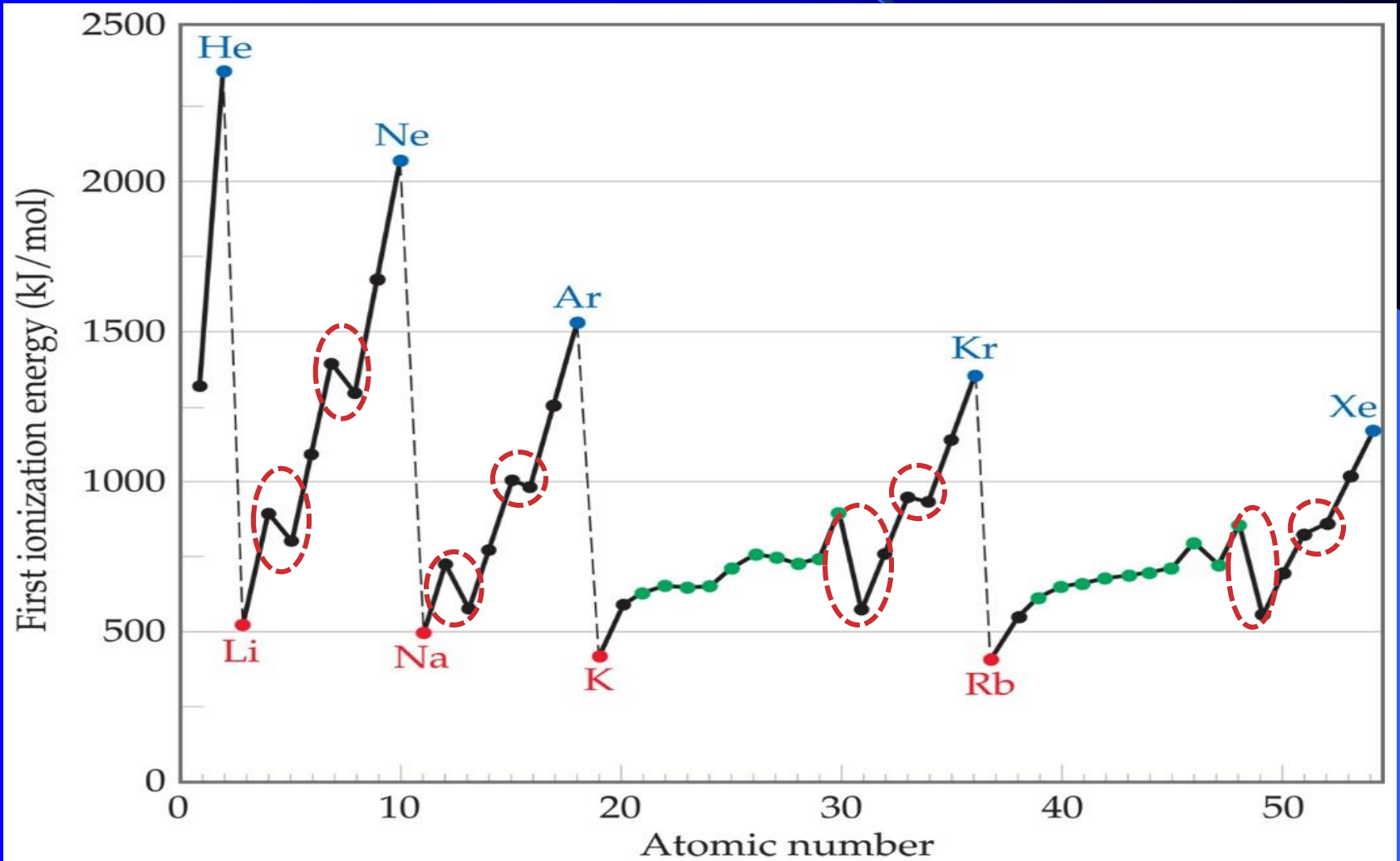
Trends in First Ionization Energies

- Generally, as one goes across a row, it gets harder to remove an electron.
 - As you go from left to right, Z_{eff} increases.



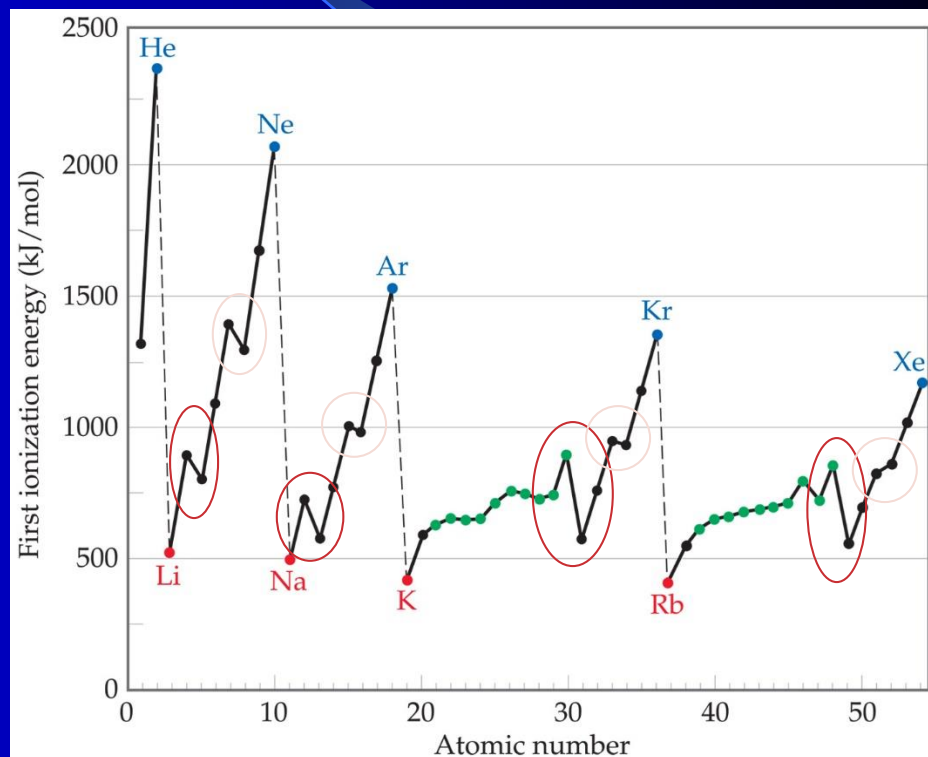
Trends in First Ionization Energies

However, there are two apparent discontinuities in this trend.



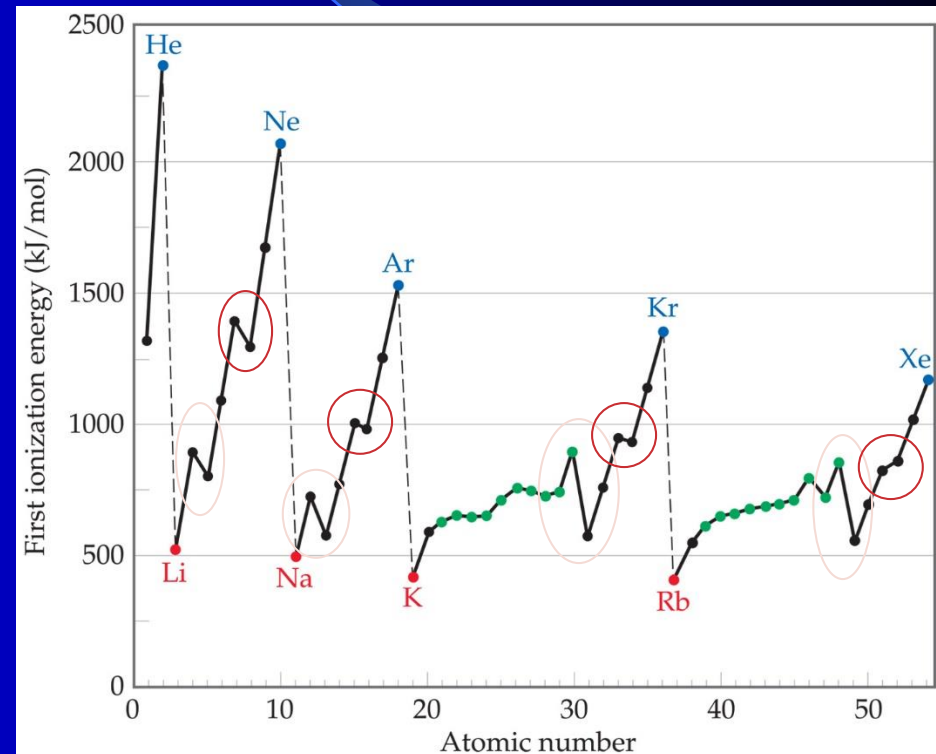
Trends in First Ionization Energies

- The first occurs between Groups IIA and IIIA.
- Electron removed from *p*-orbital rather than *s*-orbital
 - Electron farther from nucleus
 - Small amount of repulsion by *s* electrons.



Trends in First Ionization Energies

- The second occurs between Groups VA and VIA.
 - Electron removed comes from doubly occupied orbital.
 - Repulsion from other electron in orbital helps in its removal.



Electron Affinity

Energy change accompanying addition of electron to gaseous atom:



Trends in Electron Affinity

In general, electron affinity becomes more exothermic as you go from left to right across a row.

H -73						He > 0	
Li -60	Be > 0	B -27	C -122	N > 0	O -141	F -328	Ne > 0
Na -53	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ar > 0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

Trends in Electron Affinity

There are again, however, two discontinuities in this trend.

H -73							He > 0
Li -60	Be > 0	B -27	C -122	N > 0	O -141	F -328	Ne > 0
Na -53	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ar > 0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

Trends in Electron Affinity

H -73							He > 0
Li -60	Be > 0	B -27	C -122	N > 0	O -141	F -328	Ne > 0
Na -53	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ar > 0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

- The first occurs between Groups IA and IIA.
 - Added electron must go in *p*-orbital, not *s*-orbital.
 - Electron is farther from nucleus and feels repulsion from *s*-electrons.

Trends in Electron Affinity

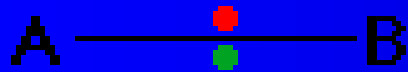
H -73							He > 0
Li -60	Be > 0	B -27	C -122	N > 0	O -141	F -328	Ne > 0
Na -53	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ar > 0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

- The second occurs between Groups IVA and VA.
 - Group VA has no empty orbitals.
 - Extra electron must go into occupied orbital, creating repulsion.

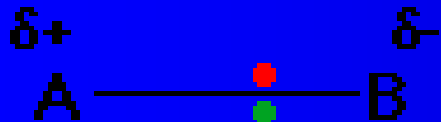
Electronegativity

- Electronegativity is a measure of the tendency of an atom to attract a bonding pair of electrons.
- The Pauling scale is the most commonly used. Fluorine (the most electronegative element) is assigned a value of 4.0, and values range down to cesium and francium which are the least electronegative at 0.7.

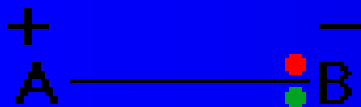
Electronegativity



Electrons are evenly attracted by atoms A and B as in H—H, Cl—Cl



Electrons are slightly more attracted To atom B than A as in H—F



Electron are totally attracted by B and atom A has lost the control of electron as in NaCl

Electronegativity

Increasing Electronegativity →

Decreasing Electronegativity ↓

1A	2A											3A	4A	5A	6A	7A
H 2.1												B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Li 1.0	Be 1.5											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
Na 0.9	Mg 1.2	3B	4B	5B	6B	7B	8B		1B	2B		Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Cs 0.8	Ba 0.9	La* 1.1	Hf 1.3	Ta 1.5	W 2.4	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9					
Fr 0.7	Ra 0.9	Ac† 1.1														

below 1.0

1.0 - 1.4

1.5 - 1.9

2.0 - 2.4

2.5 - 2.9

3.0 - 4.0

* Lanthanides: 1.1 - 1.3
† Actinides: 1.3 - 1.5